

Finite generation of canonical rings III

Christopher Hacon

University of Utah

March, 2008

Outline of the talk

1 Introduction

Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds

Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP

Outline of the talk

- 1 Introduction
 - The main Theorem
 - Surfaces
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP

Notation

- Recall that a complex projective variety $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.

Notation

- Recall that a complex projective variety $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.
- Two varieties X, X' are birational if they have isomorphic open subsets $U \cong U'$.

Notation

- Recall that a complex projective variety $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.
- Two varieties X, X' are birational if they have isomorphic open subsets $U \cong U'$.
- If X is smooth of dimension n , then T_X denotes the **tangent bundle** of X and $\omega_X = \Lambda^n T_X^\vee$ is the **canonical line bundle**.

Notation

- Recall that a complex projective variety $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.
- Two varieties X, X' are birational if they have isomorphic open subsets $U \cong U'$.
- If X is smooth of dimension n , then T_X denotes the **tangent bundle** of X and $\omega_X = \Lambda^n T_X^\vee$ is the **canonical line bundle**.
- K_X denotes (a choice of) the **canonical divisor** i.e. a formal linear combination of codimension 1 subvarieties of X such that $\omega_X = \mathcal{O}_X(K_X)$.

Notation

- Recall that a complex projective variety $X \subset \mathbb{P}_{\mathbb{C}}^N$ is defined by homogeneous polynomials $P_1, \dots, P_t \in \mathbb{C}[x_0, \dots, x_N]$.
- Two varieties X, X' are birational if they have isomorphic open subsets $U \cong U'$.
- If X is smooth of dimension n , then T_X denotes the **tangent bundle** of X and $\omega_X = \Lambda^n T_X^\vee$ is the **canonical line bundle**.
- K_X denotes (a choice of) the **canonical divisor** i.e. a formal linear combination of codimension 1 subvarieties of X such that $\omega_X = \mathcal{O}_X(K_X)$.
- We would like to use ω_X (or equivalently K_X) to understand the geometry of X .

Pluricanonical maps

- Elements of $H^0(\omega_X^{\otimes m})$ may be locally written as $f(x_1, \dots, x_n)(dx_1 \wedge \dots \wedge dx_n)^{\otimes m}$.

Pluricanonical maps

- Elements of $H^0(\omega_X^{\otimes m})$ may be locally written as $f(x_1, \dots, x_n)(dx_1 \wedge \dots \wedge dx_n)^{\otimes m}$.
- If $s_0, \dots, s_N$ is a basis of $H^0(\omega_X^{\otimes m})$, then the rational map

$$\phi_{\omega_X^{\otimes m}} : X \dashrightarrow \mathbb{P}^N$$

is defined by $x \rightarrow [s_0(x) : \dots : s_N(x)]$. (This map is not defined on the common zeroes of $s_0, \dots, s_N$.)

Pluricanonical maps

- Elements of $H^0(\omega_X^{\otimes m})$ may be locally written as $f(x_1, \dots, x_n)(dx_1 \wedge \dots \wedge dx_n)^{\otimes m}$.
- If $s_0, \dots, s_N$ is a basis of $H^0(\omega_X^{\otimes m})$, then the rational map

$$\phi_{\omega_X^{\otimes m}} : X \dashrightarrow \mathbb{P}^N$$

is defined by $x \rightarrow [s_0(x) : \dots : s_N(x)]$. (This map is not defined on the common zeroes of $s_0, \dots, s_N$.)

- If X and X' are birational, then $\forall m \geq 0$

$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$

Birational invariants

- The **canonical ring** of X is given by

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

Birational invariants

- The **canonical ring** of X is given by

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

- The **Kodaira dimension** of X is

$$\kappa(X) := \text{tr.deg.}_{\mathbb{C}} R(X) - 1 \in \{-1, 0, 1, \dots, \dim(X)\}.$$

Birational invariants

- The **canonical ring** of X is given by

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

- The **Kodaira dimension** of X is

$$\kappa(X) := \text{tr.deg.}_{\mathbb{C}} R(X) - 1 \in \{-1, 0, 1, \dots, \dim(X)\}.$$

- We have $\kappa(X) := \max_{m > 0} \{\dim \phi_{\omega_X^{\otimes m}}(X)\}.$

Birational invariants

- The **canonical ring** of X is given by

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

- The **Kodaira dimension** of X is

$$\kappa(X) := \text{tr.deg.}_{\mathbb{C}} R(X) - 1 \in \{-1, 0, 1, \dots, \dim(X)\}.$$

- We have $\kappa(X) := \max_{m > 0} \{\dim \phi_{\omega_X^{\otimes m}}(X)\}$.
- If $\kappa(X) = \dim X$, we say that X is of **general type**.

Birational invariants

- The **canonical ring** of X is given by

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

- The **Kodaira dimension** of X is

$$\kappa(X) := \text{tr.deg.}_{\mathbb{C}} R(X) - 1 \in \{-1, 0, 1, \dots, \dim(X)\}.$$

- We have $\kappa(X) := \max_{m > 0} \{\dim \phi_{\omega_X^{\otimes m}}(X)\}$.
- If $\kappa(X) = \dim X$, we say that X is of **general type**.
- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

Finite generation

Today I would like to discuss the following result:

Theorem (Birkar-Cascini-Hacon-M^cKernan and Siu)

Let X be a smooth projective variety, then $R(X)$ is finitely generated.

Finite generation

Today I would like to discuss the following result:

Theorem (Birkar-Cascini-Hacon-M^cKernan and Siu)

Let X be a smooth projective variety, then $R(X)$ is finitely generated.

The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires X to be of general type.

Curves

- Any two birational curves are isomorphic.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.
- $g = 0$: $X = \mathbb{P}^1$ and $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $R(\mathbb{P}^1) \cong \mathbb{C}$.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.
- $g = 0$: $X = \mathbb{P}^1$ and $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $R(\mathbb{P}^1) \cong \mathbb{C}$.
- $g = 1$: There is a 1-parameter family of elliptic curves;
 $K_X = 0$ (i.e. $\omega_X = \mathcal{O}_X$) so $R(X) \cong \mathbb{C}[t]$.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.
- $g = 0$: $X = \mathbb{P}^1$ and $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $R(\mathbb{P}^1) \cong \mathbb{C}$.
- $g = 1$: There is a 1-parameter family of elliptic curves;
 $K_X = 0$ (i.e. $\omega_X = \mathcal{O}_X$) so $R(X) \cong \mathbb{C}[t]$.
- $g \geq 2$: Curves of general type.
These form a $3g - 3$ -dimensional family.
 K_X is ample and $\phi_{\omega_X^{\otimes 3}}$ determines an embedding.

Curves

- Any two birational curves are isomorphic.
- Curves are topologically determined by their genus.
- $g = 0$: $X = \mathbb{P}^1$ and $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $R(\mathbb{P}^1) \cong \mathbb{C}$.
- $g = 1$: There is a 1-parameter family of elliptic curves;
 $K_X = 0$ (i.e. $\omega_X = \mathcal{O}_X$) so $R(X) \cong \mathbb{C}[t]$.
- $g \geq 2$: Curves of general type.
These form a $3g - 3$ -dimensional family.
 K_X is ample and $\phi_{\omega_X^{\otimes 3}}$ determines an embedding.
- $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$ so that $R(X)$ is finitely generated.

Birational geometry of surfaces

- The classification of surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the 20th-century.

Birational geometry of surfaces

- The classification of surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the 20th-century.
- The first problem in the birational classification of surfaces is that given any point on a surface $x \in X$, one can produce a birational morphism

$$\nu : \tilde{X} = \text{Bl}_x(X) \rightarrow X$$

known as the **blow up** of X at x .

Birational geometry of surfaces

- The classification of surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the 20th-century.
- The first problem in the birational classification of surfaces is that given any point on a surface $x \in X$, one can produce a birational morphism

$$\nu : \tilde{X} = \text{Bl}_x(X) \rightarrow X$$

known as the **blow up** of X at x .

- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.

Birational geometry of surfaces

- The classification of surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the 20th-century.
- The first problem in the birational classification of surfaces is that given any point on a surface $x \in X$, one can produce a birational morphism

$$\nu : \tilde{X} = \text{Bl}_x(X) \rightarrow X$$

known as the **blow up** of X at x .

- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.
- The exceptional curve E is called a -1 **curve** since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.

Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.
- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.

Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.
- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no -1 curves.

Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.
- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \rightarrow X_{\min}$ (given by finitely many point blow downs).

Birational geometry of surfaces III

- If $\kappa(X) = -1$, then the minimal model $X \rightarrow X_{min}$ is not unique (but it is well understood how two different minimal models are related).

Birational geometry of surfaces III

- If $\kappa(X) = -1$, then the minimal model $X \rightarrow X_{min}$ is not unique (but it is well understood how two different minimal models are related).
- If $\kappa(X) = -1$, then X is covered by rational curves (i.e. by $\mathbb{P}^1$'s). The sections of $H^0(\omega_X^{\otimes m})$ must vanish along these curves for $m > 0$. Hence $R(X) \cong \mathbb{C}$.

Birational geometry of surfaces III

- If $\kappa(X) = -1$, then the minimal model $X \rightarrow X_{\min}$ is not unique (but it is well understood how two different minimal models are related).
- If $\kappa(X) = -1$, then X is covered by rational curves (i.e. by $\mathbb{P}^1$'s). The sections of $H^0(\omega_X^{\otimes m})$ must vanish along these curves for $m > 0$. Hence $R(X) \cong \mathbb{C}$.
- If $\kappa(X) \geq 0$, then the minimal model $X \rightarrow X_{\min}$ is unique and $K_{X_{\min}}$ is nef. This means that for any curve $C \subset X_{\min}$ we have

$$\omega_{X_{\min}} \cdot C \geq 0.$$

Birational geometry of surfaces III

- If $\kappa(X) = -1$, then the minimal model $X \rightarrow X_{min}$ is not unique (but it is well understood how two different minimal models are related).
- If $\kappa(X) = -1$, then X is covered by rational curves (i.e. by $\mathbb{P}^1$'s). The sections of $H^0(\omega_X^{\otimes m})$ must vanish along these curves for $m > 0$. Hence $R(X) \cong \mathbb{C}$.
- If $\kappa(X) \geq 0$, then the minimal model $X \rightarrow X_{min}$ is unique and $K_{X_{min}}$ is nef. This means that for any curve $C \subset X_{min}$ we have

$$\omega_{X_{min}} \cdot C \geq 0.$$

- In fact one can show that $\omega_{X_{min}}$ is **semiample** i.e. there is an integer $m > 0$ such that the sections of $H^0(\omega_{X_{min}}^{\otimes m})$ have no common zeroes.

Birational geometry of surfaces IV

- Therefore, $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
 $\omega_{X_{min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1).$

Birational geometry of surfaces IV

- Therefore, $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
 $\omega_{X_{min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- It then follows that $R(X)$ is finitely generated.

Birational geometry of surfaces IV

- Therefore, $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
 $\omega_{X_{min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- It then follows that $R(X)$ is finitely generated.
- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model on which ω is positive (nef and semiample).

Birational geometry of surfaces IV

- Therefore, $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
 $\omega_{X_{min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- It then follows that $R(X)$ is finitely generated.
- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model on which ω is positive (nef and semiample).

Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
 - Threefolds
 - Running the minimal model program
- 3 Higher dimensional MMP

MMP for 3-folds

- By work of Mori, Kawamata, Kollár, Reid, Shokurov and others it is possible to generalize this picture to 3-folds:

Theorem

Assume $\dim(X) = 3$. If $\kappa(X) < 0$, then X is covered by rational curves and $R(X) \cong \mathbb{C}$.

If $\kappa(X) \geq 0$, then there is a birational map $X \dashrightarrow X_{\min}$ such that $\omega_{X_{\min}}$ is nef (i.e. $\omega_{X_{\min}} \cdot C \geq 0$ for any curve $C \subset X_{\min}$).

MMP for 3-folds

- By work of Mori, Kawamata, Kollár, Reid, Shokurov and others it is possible to generalize this picture to 3-folds:

Theorem

Assume $\dim(X) = 3$. If $\kappa(X) < 0$, then X is covered by rational curves and $R(X) \cong \mathbb{C}$.

If $\kappa(X) \geq 0$, then there is a birational map $X \dashrightarrow X_{\min}$ such that $\omega_{X_{\min}}$ is nef (i.e. $\omega_{X_{\min}} \cdot C \geq 0$ for any curve $C \subset X_{\min}$).

- More precisely, if $\kappa(X) \geq 0$, there is a sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_M = X_{\min}.$$

MMP for 3-folds

- By work of Mori, Kawamata, Kollár, Reid, Shokurov and others it is possible to generalize this picture to 3-folds:

Theorem

Assume $\dim(X) = 3$. If $\kappa(X) < 0$, then X is covered by rational curves and $R(X) \cong \mathbb{C}$.

If $\kappa(X) \geq 0$, then there is a birational map $X \dashrightarrow X_{\min}$ such that $\omega_{X_{\min}}$ is nef (i.e. $\omega_{X_{\min}} \cdot C \geq 0$ for any curve $C \subset X_{\min}$).

- More precisely, if $\kappa(X) \geq 0$, there is a sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_M = X_{\min}.$$

- It is known that $\omega_{X_{\min}}$ is semiample (i.e. $\phi_{\omega_{X_{\min}}^{\otimes m}} : X \rightarrow \mathbb{P}_{\mathbb{C}}^N$ is a morphism) and so $R(X)$ is finitely generated.

MMP for 3-folds II

- Divisorial contractions are the analog of contracting a -1 curve on a surface.

MMP for 3-folds II

- Divisorial contractions are the analog of contracting a -1 curve on a surface.
- Flips are a new operation that we will discuss below.

MMP for 3-folds II

- Divisorial contractions are the analog of contracting a -1 curve on a surface.
- Flips are a new operation that we will discuss below.
- The minimal model X_{min} is not unique.

MMP for 3-folds II

- Divisorial contractions are the analog of contracting a -1 curve on a surface.
- Flips are a new operation that we will discuss below.
- The minimal model X_{min} is not unique.
- The varieties X_i have mild singularities.

Running a MMP

- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).

Running a MMP

- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).
- We start with a projective variety X with mild singularities.

Running a MMP

- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).
- We start with a projective variety X with mild singularities.
- If ω_X is nef we are done ($X = X_{min}$).

Running a MMP

- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).
- We start with a projective variety X with mild singularities.
- If ω_X is nef we are done ($X = X_{min}$).
- If ω_X is not nef, then consider the **cone of effective curves**

$$\overline{NE}(X) = \left\{ \sum r_i C_i \mid r_i \in \mathbb{R}^{\geq 0} \right\} / \equiv$$

(here $C \equiv C'$ or $[C] = [C']$ if for any line bundle L on X , we have $L \cdot (C - C') = 0$).

Running a MMP

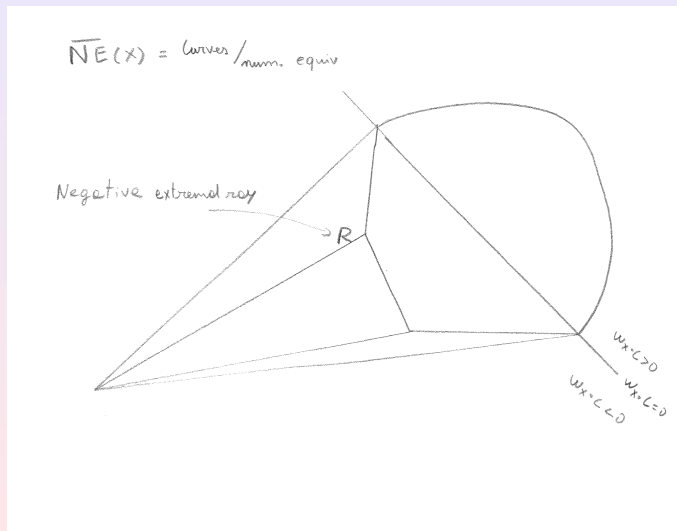
- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).
- We start with a projective variety X with mild singularities.
- If ω_X is nef we are done ($X = X_{min}$).
- If ω_X is not nef, then consider the **cone of effective curves**

$$\overline{NE}(X) = \left\{ \sum r_i C_i \mid r_i \in \mathbb{R}^{\geq 0} \right\} / \equiv$$

(here $C \equiv C'$ or $[C] = [C']$ if for any line bundle L on X , we have $L \cdot (C - C') = 0$).

- By the Cone Theorem, there is a negative extremal ray $R = \mathbb{R}^+[C]$ where $C \subset X$ is a curve such that $\omega_X \cdot C < 0$ and there is a morphism $\text{Cont}_R : X \rightarrow Z$ such that a curve $D \subset X$ is contracted if and only if $[D] \in R^+[C]$.

The Cone Theorem



Running a MMP II

- If $\dim Z < \dim X$, then X is covered by ω_X negative rational curves. (This only happens if $\kappa(X) < 0$.)

Running a MMP II

- If $\dim Z < \dim X$, then X is covered by ω_X negative rational curves. (This only happens if $\kappa(X) < 0$.)
- If $\dim Z = \dim X$ and Cont_R is **divisorial** (i.e. $\dim \text{Exc}(X \rightarrow Z) = \dim X - 1$) then Z has mild singularities and we may replace X by Z (and restart the whole process).

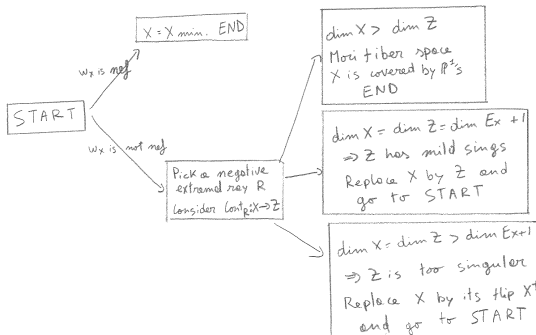
Running a MMP II

- If $\dim Z < \dim X$, then X is covered by ω_X negative rational curves. (This only happens if $\kappa(X) < 0$.)
- If $\dim Z = \dim X$ and Cont_R is **divisorial** (i.e. $\dim \text{Exc}(X \rightarrow Z) = \dim X - 1$) then Z has mild singularities and we may replace X by Z (and restart the whole process).
- If $\dim Z = \dim X$ and Cont_R is **small** (i.e. $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$) then Z has “bad” singularities. In particular $\omega_Z \cdot C$ does not always make sense.

Running a MMP II

- If $\dim Z < \dim X$, then X is covered by ω_X negative rational curves. (This only happens if $\kappa(X) < 0$.)
- If $\dim Z = \dim X$ and Cont_R is **divisorial** (i.e. $\dim \text{Exc}(X \rightarrow Z) = \dim X - 1$) then Z has mild singularities and we may replace X by Z (and restart the whole process).
- If $\dim Z = \dim X$ and Cont_R is **small** (i.e. $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$) then Z has “bad” singularities. In particular $\omega_Z \cdot C$ does not always make sense.
- Instead of replacing X by Z , we will replace X by its flip X^+ .

MMP flow chart



Flips

- The **flip** of $X \rightarrow Z$ is a small birational morphism $X^+ \rightarrow Z$ such that
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
 - 2) X^+ has mild singularities, and
 - 3) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.

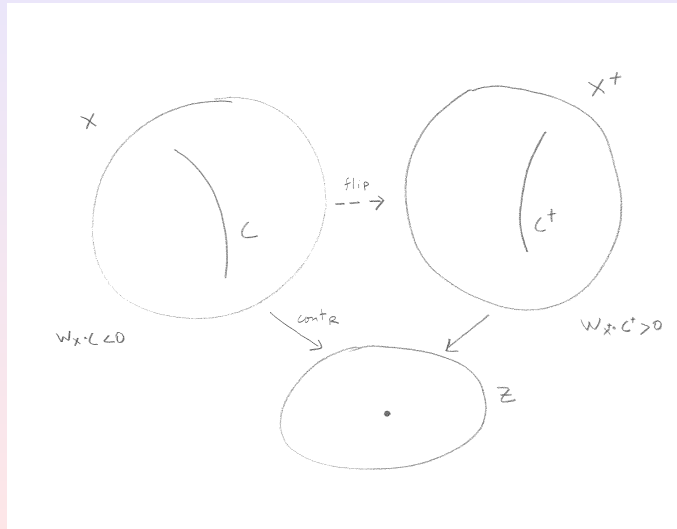
Flips

- The **flip** of $X \rightarrow Z$ is a small birational morphism $X^+ \rightarrow Z$ such that
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
 - 2) X^+ has mild singularities, and
 - 3) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.
- In other words, the flip $X \dashrightarrow X^+$ is a surgery that replaces ω_X negative curves by ω_{X^+} positive curves.

Flips

- The **flip** of $X \rightarrow Z$ is a small birational morphism $X^+ \rightarrow Z$ such that
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
 - 2) X^+ has mild singularities, and
 - 3) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.
- In other words, the flip $X \dashrightarrow X^+$ is a surgery that replaces ω_X negative curves by ω_{X^+} positive curves.
- If the flip of $X \rightarrow Z$ exists, it is unique and given by $\text{Proj}_Z R(X)$. So that to construct the flip we must show that $R(X)$ is finitely generated over Z (i.e. if $Z = \text{Spec} A$, $R(X)$ is a finitely generated A -algebra).

Picture of a flip



Flips II

- In dimension 3, flips were constructed by Mori.

Flips II

- In dimension 3, flips were constructed by Mori.
- In order to complete the MMP, one must show that there are no infinite sequence of flips and divisorial contractions.

Flips II

- In dimension 3, flips were constructed by Mori.
- In order to complete the MMP, one must show that there are no infinite sequence of flips and divisorial contractions.
- If $X \rightarrow Z$ is a divisorial contraction, then $\rho(Z) = \rho(X) - 1 \geq 0$. If $X \dashrightarrow X^+$ is a flip, then $\rho(X^+) = \rho(X)$. Therefore, it suffices to show that there is no infinite sequence of flips.

Flips II

- In dimension 3, flips were constructed by Mori.
- In order to complete the MMP, one must show that there are no infinite sequence of flips and divisorial contractions.
- If $X \rightarrow Z$ is a divisorial contraction, then $\rho(Z) = \rho(X) - 1 \geq 0$. If $X \dashrightarrow X^+$ is a flip, then $\rho(X^+) = \rho(X)$. Therefore, it suffices to show that there is no infinite sequence of flips.
- In dimension 3, this is relatively easy to do.

Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP
 - Reduction to log pairs of general type
 - Idea of the proof

Log pairs

- It is necessary to work in greater generality with Kawamata log terminal (KLT) log-pairs (X, Δ) .

Log pairs

- It is necessary to work in greater generality with Kawamata log terminal (KLT) log-pairs (X, Δ) .
- One should think of these pairs as representing varieties with mild singularities.

Log pairs

- It is necessary to work in greater generality with Kawamata log terminal (KLT) log-pairs (X, Δ) .
- One should think of these pairs as representing varieties with mild singularities.
- Roughly speaking, we may assume that X is smooth and $\Delta = \sum \delta_i \Delta_i$ is a formal sum of smooth codimension 1 subvarieties meeting transversely with coefficients $0 \leq \delta_i < 1$.

Log pairs

- It is necessary to work in greater generality with Kawamata log terminal (KLT) log-pairs (X, Δ) .
- One should think of these pairs as representing varieties with mild singularities.
- Roughly speaking, we may assume that X is smooth and $\Delta = \sum \delta_i \Delta_i$ is a formal sum of smooth codimension 1 subvarieties meeting transversely with coefficients $0 \leq \delta_i < 1$.
- The proof proceeds by induction on the dimension of X . We relate ω_X (equivalently K_X) to its restriction to a divisor S (i.e. a codimension 1 subvariety which typically belongs to $|mK_X|$).

Log pairs

- It is necessary to work in greater generality with Kawamata log terminal (KLT) log-pairs (X, Δ) .
- One should think of these pairs as representing varieties with mild singularities.
- Roughly speaking, we may assume that X is smooth and $\Delta = \sum \delta_i \Delta_i$ is a formal sum of smooth codimension 1 subvarieties meeting transversely with coefficients $0 \leq \delta_i < 1$.
- The proof proceeds by induction on the dimension of X . We relate ω_X (equivalently K_X) to its restriction to a divisor S (i.e. a codimension 1 subvariety which typically belongs to $|mK_X|$).
- This is achieved by the adjunction formula which has the form $(K_X + S)|_S = K_S + \Delta_S$ for some $\mathbb{Q}$ -divisor Δ_S .

A Theorem of Fujino-Mori

Mori and Fujino have shown that:

Theorem

Let (X, Δ) be any KLT $\mathbb{Q}$ -factorial pair with $\Delta \in \text{Div}_{\mathbb{Q}}(X)$. Then there exists a KLT $\mathbb{Q}$ -factorial pair (Y, Γ) of general type with $\Gamma \in \text{Div}_{\mathbb{Q}}(Y)$ such that

$$R(K_X + \Delta)^{(m)} \cong R(K_Y + \Gamma)^{(m)}$$

for any sufficiently divisible integer $m > 0$.

Recall that $K_Y + \Gamma$ is of general type if $\text{tr.deg.}_{\mathbb{C}} R(K_Y + \Gamma)^{(m)} - 1 = \dim Y$.

A Theorem of Fujino-Mori II

- It follows that to show the main theorem (i.e. that $R(X)$ is finitely generated), it suffices to prove the corresponding statement for KLT log pairs of general type.

A Theorem of Fujino-Mori II

- It follows that to show the main theorem (i.e. that $R(X)$ is finitely generated), it suffices to prove the corresponding statement for KLT log pairs of general type.
- If (X, Δ) is KLT and $K_X + \Delta$ is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov, $K_X + \Delta$ is semiample.

A Theorem of Fujino-Mori II

- It follows that to show the main theorem (i.e. that $R(X)$ is finitely generated), it suffices to prove the corresponding statement for KLT log pairs of general type.
- If (X, Δ) is KLT and $K_X + \Delta$ is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov, $K_X + \Delta$ is semiample.
- As we have mentioned before, this implies that $R(K_X + \Delta)$ is finitely generated.

A Theorem of Fujino-Mori II

- It follows that to show the main theorem (i.e. that $R(X)$ is finitely generated), it suffices to prove the corresponding statement for KLT log pairs of general type.
- If (X, Δ) is KLT and $K_X + \Delta$ is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov, $K_X + \Delta$ is semiample.
- As we have mentioned before, this implies that $R(K_X + \Delta)$ is finitely generated.
- Therefore, to show that $R(X)$ is finitely generated (for any smooth complex projective variety), it suffices to show that minimal models exist for KLT pairs of general type.

Existence of minimal models for KLT pairs of general type.

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let (X, Δ) be any KLT pair of general type. Then there is a minimal model $f : X \dashrightarrow X_{\min}$ (i.e. $K_{X_{\min}} + f_\Delta$ is semiample).*

Existence of minimal models for KLT pairs of general type.

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let (X, Δ) be any KLT pair of general type. Then there is a minimal model $f : X \dashrightarrow X_{\min}$ (i.e. $K_{X_{\min}} + f_\Delta$ is semiample).*

Corollary

Flips exist (for KLT pairs; in any dimension).

Existence of minimal models for KLT pairs of general type.

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let (X, Δ) be any KLT pair of general type. Then there is a minimal model $f : X \dashrightarrow X_{\min}$ (i.e. $K_{X_{\min}} + f_\Delta$ is semiample).*

Corollary

Flips exist (for KLT pairs; in any dimension).

Remark: We are unable to show termination of flips. But we can pick a sequence of flips that terminates.

The induction

- The main idea is to proceed by induction on the dimension.
We show that:

The induction

- The main idea is to proceed by induction on the dimension.
We show that:
- (Hacon-M^cKernan) If minimal models (for KLT pairs of general type) exist in dimension n , then flips exist in dimension $n + 1$. (Uses ideas of Shokurov and Siu-Kawamata-Tsuji.)

The induction

- The main idea is to proceed by induction on the dimension.
We show that:
- (Hacon-M^cKernan) If minimal models (for KLT pairs of general type) exist in dimension n , then flips exist in dimension $n + 1$. (Uses ideas of Shokurov and Siu-Kawamata-Tsuji.)
- (Birkar-Cascini-Hacon-M^cKernan) If flips exist in dimension $n + 1$ (and minimal models exist in dimension n) then minimal models exist in dimension $n + 1$. (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

Existence of flips

- By a result of Shokurov, the existence of flips is reduced to the existence of PI-flips.

Existence of flips

- By a result of Shokurov, the existence of flips is reduced to the existence of PI-flips.
- This means that we may assume that (X, Δ) is PLT i.e. $\Delta = S + \sum \delta_i \Delta_i$ with $0 \leq \delta_i < 1$ and S is the unique center along which (X, Δ) is not KLT.

Existence of flips

- By a result of Shokurov, the existence of flips is reduced to the existence of PL-flips.
- This means that we may assume that (X, Δ) is PLT i.e. $\Delta = S + \sum \delta_i \Delta_i$ with $0 \leq \delta_i < 1$ and S is the unique center along which (X, Δ) is not KLT.
- By adjunction, we may write $(K_X + \Delta)|_S = K_S + \Delta_S$ where (S, Δ_S) is KLT.
- Shokurov's observation is that to show that $R(K_X + \Delta)$ is finitely generated over Z (and hence that the flip exists), it suffices to show that the restricted algebra

$$R_S(K_X + \Delta) := \text{Im}(R(K_X + \Delta) \rightarrow R(K_S + \Delta_S))$$

is finitely generated (over Z).

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.
- This is too much to expect. However, we show that $R_S(K_X + \Delta) = R(K_S + \Theta)$ where $0 \leq \Theta \leq \Delta_S$ and (S, Θ) is KLT.

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.
- This is too much to expect. However, we show that $R_S(K_X + \Delta) = R(K_S + \Theta)$ where $0 \leq \Theta \leq \Delta_S$ and (S, Θ) is KLT.
- The divisor Θ is the limit of $\mathbb{Q}$ -divisors Θ_m defined by subtracting common components of Δ_S and $\text{Bs}(m(K_X + \Delta))/m$ from Δ_S .

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.
- This is too much to expect. However, we show that $R_S(K_X + \Delta) = R(K_S + \Theta)$ where $0 \leq \Theta \leq \Delta_S$ and (S, Θ) is KLT.
- The divisor Θ is the limit of $\mathbb{Q}$ -divisors Θ_m defined by subtracting common components of Δ_S and $\text{Bs}(m(K_X + \Delta))/m$ from Δ_S .
- Using ideas of Siu, Kawamata and Tsuji we show that $H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S)))$.

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.
- This is too much to expect. However, we show that $R_S(K_X + \Delta) = R(K_S + \Theta)$ where $0 \leq \Theta \leq \Delta_S$ and (S, Θ) is KLT.
- The divisor Θ is the limit of $\mathbb{Q}$ -divisors Θ_m defined by subtracting common components of Δ_S and $\text{Bs}(m(K_X + \Delta))/m$ from Δ_S .
- Using ideas of Siu, Kawamata and Tsuji we show that $H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S)))$.
- The divisor Θ is a priori an $\mathbb{R}$ -divisor.

Existence of flips II

- If $R_S(K_X + \Delta) = R(K_S + \Delta_S)$ then we are done as we are assuming that minimal models exist in dimension $n = \dim S$.
- This is too much to expect. However, we show that $R_S(K_X + \Delta) = R(K_S + \Theta)$ where $0 \leq \Theta \leq \Delta_S$ and (S, Θ) is KLT.
- The divisor Θ is the limit of $\mathbb{Q}$ -divisors Θ_m defined by subtracting common components of Δ_S and $\text{Bs}(m(K_X + \Delta))/m$ from Δ_S .
- Using ideas of Siu, Kawamata and Tsuji we show that
$$H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S))).$$
- The divisor Θ is a priori an $\mathbb{R}$ -divisor.
- Using a Diophantine approximation argument, we show that Θ is in fact a $\mathbb{Q}$ -divisor and $\Theta = \Theta_m$ for some $m > 0$.

The MMP with scaling

- In order to prove the existence of minimal models (given the existence of flips), instead of trying to show that any sequence of flips terminates, we carefully choose our sequence of flips (and divisorial contractions).

The MMP with scaling

- In order to prove the existence of minimal models (given the existence of flips), instead of trying to show that any sequence of flips terminates, we carefully choose our sequence of flips (and divisorial contractions).
- If (X, Δ) is a KLT pair, we pick an ample divisor A such that $K_X + \Delta + A$ is ample (and hence X is the minimal model of $(X, \Delta + A)$).

The MMP with scaling

- In order to prove the existence of minimal models (given the existence of flips), instead of trying to show that any sequence of flips terminates, we carefully choose our sequence of flips (and divisorial contractions).
- If (X, Δ) is a KLT pair, we pick an ample divisor A such that $K_X + \Delta + A$ is ample (and hence X is the minimal model of $(X, \Delta + A)$).
- We then consider $t_1 = \inf\{t \geq 0 \mid K_X + \Delta + tA \text{ is nef}\}$.

The MMP with scaling

- In order to prove the existence of minimal models (given the existence of flips), instead of trying to show that any sequence of flips terminates, we carefully choose our sequence of flips (and divisorial contractions).
- If (X, Δ) is a KLT pair, we pick an ample divisor A such that $K_X + \Delta + A$ is ample (and hence X is the minimal model of $(X, \Delta + A)$).
- We then consider $t_1 = \inf\{t \geq 0 \mid K_X + \Delta + tA \text{ is nef}\}$.
- If $t_1 = 0$ we are done. Otherwise we flip (resp. perform a divisorial contraction) along a ray $R = \mathbb{R}^+[C]$ where $(K_X + \Delta + t_1 A) \cdot C = 0$.

The MMP with scaling

- In order to prove the existence of minimal models (given the existence of flips), instead of trying to show that any sequence of flips terminates, we carefully choose our sequence of flips (and divisorial contractions).
- If (X, Δ) is a KLT pair, we pick an ample divisor A such that $K_X + \Delta + A$ is ample (and hence X is the minimal model of $(X, \Delta + A)$).
- We then consider $t_1 = \inf\{t \geq 0 \mid K_X + \Delta + tA \text{ is nef}\}$.
- If $t_1 = 0$ we are done. Otherwise we flip (resp. perform a divisorial contraction) along a ray $R = \mathbb{R}^+[C]$ where $(K_X + \Delta + t_1 A) \cdot C = 0$.
- In this way, we obtain a sequence of rational numbers $1 \geq t_1 \geq t_2 \cdots \geq 0$ and of flips/div. contractions $X_i \dashrightarrow X_{i+1}$ such that X_i is a minimal model for $(X, \Delta + t_i A)$.

The MMP with scaling II

- Using ideas of Shokurov, we show that the set of all minimal models for pairs of the form $(X, \Delta + tA)$ with $t \in [0, 1]$ is finite.

The MMP with scaling II

- Using ideas of Shokurov, we show that the set of all minimal models for pairs of the form $(X, \Delta + tA)$ with $t \in [0, 1]$ is finite.
- This implies that the sequence $\dots X_i \dashrightarrow X_{i+1} \dots$ is not infinite and so the above process produces the required minimal model.

The MMP with scaling II

- Using ideas of Shokurov, we show that the set of all minimal models for pairs of the form $(X, \Delta + tA)$ with $t \in [0, 1]$ is finite.
- This implies that the sequence $\dots X_i \dashrightarrow X_{i+1} \dots$ is not infinite and so the above process produces the required minimal model.
- Note that to prove the above result on the finiteness of minimal models, we must use a compactness argument and hence it is critical to work with $\mathbb{R}$ -divisors.